Rajdeep Gill

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Education

University of Manitoba May 2026

Bachelor of Science in Computer Engineering **4.0 GPA**

Experience

Robinhood **Sept 2025 - Present**

Software Engineer Contractor *Toronto, ON*

- Contracted to develop advanced margin and equities trading tools for active traders, enhancing risk management capabilities and decision-making through sophisticated web-based trading interfaces.

Robinhood **May 2025 - Sept 2025**

Software Engineer Intern *Toronto, ON*

- Reduced customer margin calls by **37%** and boosted margin trading activity by **23%** by launching a real-time maintenance dashboard with embedded trade execution, surfacing key risk indicators like maintenance requirement.
- Architected a cross-platform margin experience using a server-driven UI approach to unify business logic across Web, iOS, and Android, driving a **43%** increase in margin usage.
- Accelerated the adoption of a new **Protobuf design system** by implementing its foundational components, adding a logging framework to improve observability, and resolving initial implementation gaps.
- Repaired the core integration suite for the margin experience and added **20+ test suites**, achieving **86%** coverage.

Biosensors Research Lab **May 2024 - Dec 2024**

Undergraduate Research Student *Winnipeg, MB*

- Achieved **85%** accuracy in predicting cell viability from **10,000+** diffraction images using a **custom CNN**, establishing a viable, non-invasive measurement method for cell health
- Built an interactive model validation GUI in Python that rendered SHAP values to provide real-time explanations of prediction drivers, enabling researchers to debug model behavior on a per-cell basis
- Architected an end-to-end data processing pipeline that boosted morphological parameter extraction accuracy by **17%** and cut pre-processing runtime by **50%** through parallelization using Python's multiprocessing library
- Replaced a high-risk system of manually-shared hard drives with a centralized MySQL database, eliminating data loss and versioning conflicts, and enabling complex, cross-experiment queries for the first time.

Biosensors Research Lab **May 2023 - Aug 2023**

Undergraduate Research Student *Winnipeg, MB*

- Engineered a general-purpose Python control framework for the lab's oscilloscope, abstracting low-level hardware commands into a high-level API that empowered researchers to script and automate entire experimental sequences.
- Slashed data analysis runtime by over **90%** (1 hour to <6 minutes) by refactoring legacy Python scripts, replacing inefficient loops with vectorized NumPy operations
- Developed a novel classification system using a Gaussian Mixture Model to determine cell viability from time-series impedance data, creating a probabilistic tool for assessing sample health

Projects

Void - AI App Builder *(Next.js, TypeScript, Prisma, Docker)*

- Developed an app to generate and deploy web apps from user prompts, integrating a usage-based Stripe billing system to monetize user compute time and API token consumption
- Implemented a persistent background job queue to process long-running AI build tasks asynchronously, allowing users to submit jobs and receive the completed result at a later time, even after closing the browser
- Utilized Docker for user code sandboxing, secure code execution and real-time application previews

Collaborative Document Editor *(React, TypeScript, Socket.io, Flask)*

- Engineered a collaborative text editor for up to 20 simultaneous users by broadcasting text operations, cursor locations, and online statuses, via WebSockets, achieving a sub-250ms synchronization latency
- Integrated Gemini to provide real-time suggestions for text, formatting and content generation, reducing writing time

Skills

Programming Languages: C++, Python, TypeScript, JavaScript, Java

Frameworks and Libraries: React, NextJS, PyTorch, OpenMPI, Node.js, Express, Flask

Tools: Git, Docker, PostgreSQL, Github Actions, Bazel, gRPC, Protocol Buffers (Protobuf)